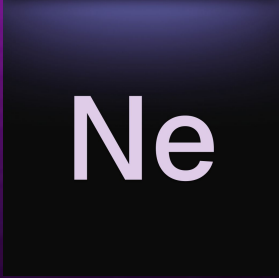


Low Level C++ Theory & Applications

Amlal El Mahrouss

I'm Amlal! Nice to meet you all!

The logo consists of a dark blue square with a subtle gradient and a slight shadow. Inside the square, the letters 'Ne' are written in a white, bold, sans-serif font.

Ne

Email: amlal@ne-app.eu

Blog: <https://amlal.nekernel.org>

Purpose.

→ In this talk we're going to layout techniques and decisions to take in order to successfully develop a low-level system in modern C++.

The standards involved:

→ C++17 and later.

I: Principles First.

Amlal El Mahrouss

Observation: What we will NOT cover.

- Exact result of the defined finite number $\text{TREE}(3)$ in the Kruskal's tree theorem.
- $P \stackrel{?}{=} NP$ in Theoretical Computer Science.
- Tabs vs Spaces. Or Emacs vs Neovim.

II: Common RAII.

Amlal El Mahrouss

Observation: construct, use, destruct:

→ `std::unique_ptr`.

→ `std::shared_ptr`.

→ `std::weak_ptr`.

→ `std::unique_lock`

III: Common Patterns.

Amlal El Mahrouss

IV: Applications and Examples.

Amlal El Mahrouss

IV: Applications and Examples.

Amlal El Mahrouss

V: Final Remarks.

Amlal El Mahrouss

VI: Conclusion.

Amlal El Mahrouss